



Real World Planning and Acting

Serializable Subgoals and Hierarchical Planning Notes

Example: Sussman Anomaly

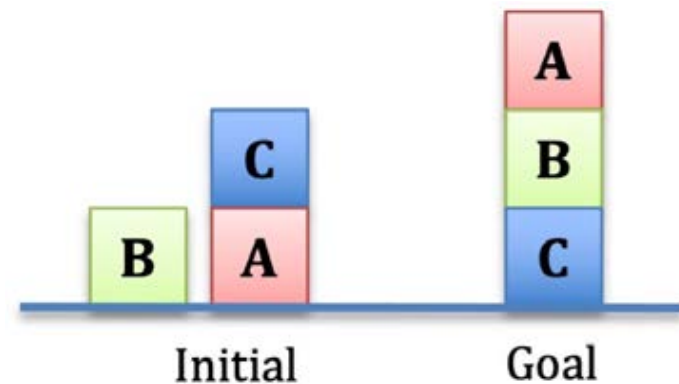
Consider three blocks A, B, and C with the following:

- **Initial State:**

- Block C is on Block A ($\text{on}(\text{C}, \text{A})$).
- Block A is on the table.
- Block B is on the table ($\text{onTable}(\text{B})$).

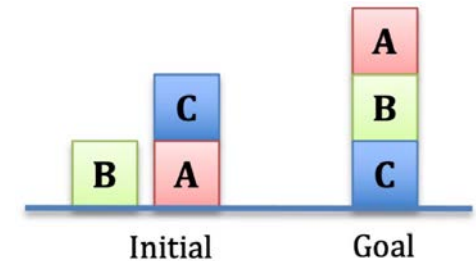
- **Goal State:**

- Block A on Block B ($\text{on}(\text{A}, \text{B})$).
- Block B on Block C ($\text{on}(\text{B}, \text{C})$).



Original source: Sussman, G. J. (1975). A Computer Model of Skill Acquisition. Elsevier.

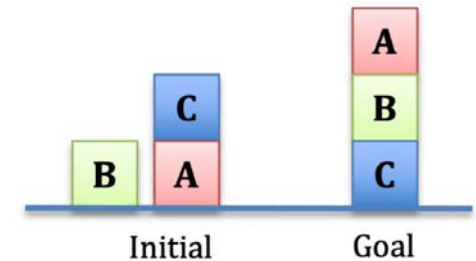
Example: Sussman Anomaly



Sequence 1: Achieving $\text{on}(A, B)$ first, then $\text{on}(B, C)$

1. $\text{Move}(\text{table}, C, A)$ – Move Block C from Block A to the table.
2. $\text{Move}(A, \text{table}, B)$ – Move Block A from the table to Block B.
 - Subgoal Achieved: $\text{on}(A, B)$.
3. $\text{Move}(\text{table}, A, B)$ – **Undo** the subgoal $\text{on}(A, B)$ by moving Block A from Block B back to the table to clear B.
4. $\text{Move}(B, \text{table}, C)$ – Move Block B from the table to Block C.
 - Subgoal Achieved: $\text{on}(B, C)$.
5. $\text{Move}(A, \text{table}, B)$ – Finally, move Block A from the table to Block B again.
 - Final State: $\text{on}(A, B)$ and $\text{on}(B, C)$ achieved.

Example: Sussman Anomaly



Sequence 2: Achieving $\text{on}(B, C)$ first, then $\text{on}(A, B)$

1. $\text{Move}(B, \text{table}, C)$ – Move Block B from the table to Block C.
 - Subgoal Achieved: $\text{on}(B, C)$.
2. $\text{Movetotable}(B, C)$ – **Undo** the subgoal $\text{on}(B, C)$ by moving Block B from Block C back to the table to clear C.
3. $\text{Movetotable}(C, A)$ – Move Block C from Block A to the table.
4. $\text{Move}(A, \text{table}, B)$ – Move Block A from the table to Block B.
 - Subgoal Achieved: $\text{on}(A, B)$.
5. $\text{Movetotable}(A, B)$ – **Undo** the subgoal $\text{on}(A, B)$ by moving Block A from Block B back to the table to clear B.
6. $\text{Move}(B, \text{table}, C)$ – Move Block B from the table to Block C again.
 - Subgoal Achieved: $\text{on}(B, C)$.
7. $\text{Move}(A, \text{table}, B)$ – Finally, move Block A from the table to Block B again.
 - Final State: $\text{on}(A, B)$ and $\text{on}(B, C)$ achieved.

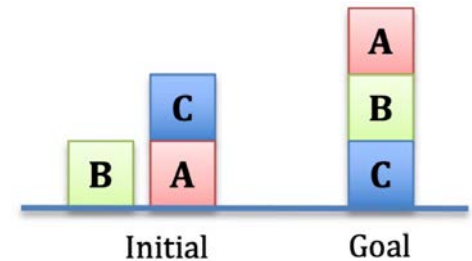


Example: Sussman Anomaly

Observations:

- In both sequences, achieving one goal required undoing previously achieved subgoals, which exemplifies Sussman Anomaly. Specifically:
 - In **Sequence 1**, the subgoal $\text{on}(A, B)$ was undone in Step 3 to achieve $\text{on}(B, C)$.
 - In **Sequence 2**, the subgoal $\text{on}(B, C)$ was undone in Step 2, and $\text{on}(A, B)$ was undone in Step 5 to rearrange the blocks.
- These steps highlight the interdependencies and conflicts between subgoals, demonstrating the complexity and inefficiency in achieving all goals without having to backtrack and redo certain actions.

Example: Sussman Anomaly



- What is the problem with independent subgoals?
 - Moving two blocks to specific positions may require a sequence that undoes one goal to achieve another.
- **Serializable Subgoals:**
 - A planning approach where subgoals are achieved in a specific sequence without conflict.
 - Ensures each subgoal is completed without undoing previously achieved goals.
 - Helps avoid issues like Sussman Anomaly.

Example: Air Cargo Transportation



- Original problem:
 - 10 airports, 50 planes and 200 cargos
 - Total no. of states = $10^{50} \times (50 + 10)^{200} \approx 10^{405}$
- Assumption:
 - All cargos are just in 5 airports, instead of 10, and all cargos in the same airport have the same destination.
- Reformulation:
 - Drop all the irrelevant *At* fluents (What are the relevant ones?)
 - Total no. of states = $10^5 \times (5 + 10)^5 \approx 10^{11}$
- Solution:
 - Shorter than that for original problem (admissible heuristic)
 - Extensible by adding relevant actions

Example: Original Problem Setup



- Initial Problem Setup:

- 10 airports: Each airport can be a possible location for planes and cargo.
- 50 planes: Each plane can be at one of the 10 airports.
- 200 pieces of cargo: Each piece of cargo can either be on one of the planes or unloaded at one of the airports.

- Total Number of States:

1. Planes' locations:

- Each of the 50 planes can be at any of the 10 airports.
- The total number of combinations for plane locations is 10^{50}

2. Cargo locations:

- Each of the 200 pieces of cargo can either be:
 - On one of the 50 planes (50 options), or
 - At one of the 10 airports (10 options).
- Therefore, each piece of cargo has 60 possible locations (50 planes + 10 airports).
- The total number of combinations for cargo locations is $(50 + 10)^{200}$

3. Total number of states:

- Multiply the number of possible plane locations by the number of possible cargo locations:

$$\text{Total states} = 10^{50} \times (50 + 10)^{200} \approx 10^{405}$$

Example: Abstracted Problem Setup



- Now, consider that:
 - All packages are at just 5 of the airports.
 - All packages at a given airport have the same destination.
- This allows us to make significant abstractions:
 1. Planes' locations:
 - We abstract the problem by considering only 5 "big" planes, each representing a group of planes heading to the same destination.
 - These 5 planes can be at any of the 10 airports.
 - The total number of combinations for these 5 planes is 10^5 .
 2. Cargo locations:
 - We abstract the problem by considering 5 "big" packages, each representing all the cargo at one airport.
 - Each "big" package can either be:
 - On one of the 5 "big" planes (5 options), or
 - At one of the 10 airports (10 options).
 - Therefore, each "big" package has 15 possible locations (5 planes + 10 airports).
 - The total number of combinations for the 5 "big" packages is $(5 + 10)^5$.
 3. Total number of abstracted states:
 - Multiply the number of possible "big" plane locations by the number of possible "big" package locations:

$$\text{Total abstracted states} = 10^5 \times (5 + 10)^5 \approx 10^{11}$$

Example: Hierarchical Planning Complexity

- Goal: Get from Home to SIN Airport
 - Primitive actions: walk, open door, get in taxi, pay, etc
- Flat planner (non-hierarchical):
 - Branching factor $b \approx 10$ primitive actions/step
 - Optimal plan length $d = 30$ primitive steps
 - Cost: $O(b^d) = O(10^{30}) \rightarrow$ astronomically large
- HTN planner (illustrative):
 - HLA: Go(Home, SIN)
 - Refinements: $r = 3$ (e.g., Drive / Taxi / MRT)
 - Each refinement expands to about $k = 10$ primitive actions
 - No. of refinement steps $\approx \frac{d-1}{k-1} = \frac{29}{9} \approx 3.22$
 - Cost (approx.): $O(r^{(d-1)/(k-1)}) = O(3^{3.22}) \approx 34 \rightarrow$ Huge exponential savings!

Note:

- d = total primitive actions in the final plan.
- $(d-1)/(k-1)$ arises because each refinement adds $k - 1$ tasks; to reach d tasks we need $d-1$ additions.